

**TASK 05-50-13-200-801-A****EFFECTIVITY: ALL****1. Landing Gear Down Overspeed - Inspection/Check****A. General**

- (1) This section gives the procedure you must do when at least one of the events below occurs:
- Retraction of the landing gear at an airspeed higher than the airspeed limit for landing gear operation.
 - Extension of the landing gear at an airspeed higher than the airspeed limit for landing gear operation.
 - Flight with the landing gear down and locked at an airspeed higher than the airspeed limit for flight with landing gear down.
- (2) This maintenance technician is necessary to do this task:
- 1 - Mechanical Systems
- (3) This maintenance technician is necessary to do this task:
- 1 - Electrical/Avionics Systems

B. References

REFERENCE	DESIGNATION
AMM TASK 32-00-00-860-801-A/200	Landing Gear - Prepare for Maintenance
AMM TASK 32-00-00-860-802-A/200	Landing Gear - Restore after Maintenance
AMM TASK 32-12-00-820-801-A/200	Main-Landing-Gear Doors - Adjustment
AMM TASK 32-22-01-820-801-A/200	Nose-Landing-Gear Forward-Door Rods Linkage - Adjustment
AMM TASK 32-22-02-860-801-A/200	Nose-Landing-Gear Forward Door (open) - Configuration
AMM TASK 32-22-02-860-802-A/200	Nose-Landing-Gear Forward Door (close) - Configuration
AMM TASK 32-22-03-820-801-A/200	Nose-Landing-Gear Aft-Door - Detailed Dimensional Inspection
AMM TASK 32-33-00-710-801-A/500	Landing-Gear Extension-and-Retraction System (On Jacks) - Operational Test
AMM TASK 32-34-00-710-801-A/500	Emergency-Extension System (On Jacks) - Operational Test

C. Overspeed Landing-Gear Operation - Event Assessment**SUBTASK 200-002-A**

- (1) The IAS limits for flight with landing-gear in transit or locked down are given in Section 2 of the AFM and named as follow:
- V_{LO} (e) is the maximum speed that the landing gear can be safely extended.
 - V_{LO} (r) is the maximum speed that the landing gear can be safely retracted.



- V_{LE} is the maximum speed that the airplane can be safely flown with the landing gear down and locked.
- (2) The maximum value speed is as follows to give the operational envelope for flight with the landing-gear in transit or locked down:
- (a) The V_{LO} (e) value is 265 KIAS.
 - (b) The V_{LO} (r) value is 235 KIAS.
 - (c) The V_{LE} value is 265 KIAS.
- (3) When you consider the operational envelope, two threshold values are set to classify the severity of the event. The severity of the event will lead to one of these situations:
- IAS below the Low-Threshold (T_{LOW}):
The T_{LOW} value defines an excess to the operational envelope in a degree of severity that no inspection of the aircraft is necessary if the event occurred only once during the flight period and did not last more than 20 (twenty) seconds.
 - IAS between the Low-Threshold (T_{LOW}) and High-Threshold (T_{HIGH}):
Defines a situation clearly out of the aircraft envelope, but the operator can still restore the aircraft airworthiness, without EMBRAER involvement, if no discrepancies are found during the applicable inspections.
 - IAS above the High-Threshold (T_{HIGH}):
The T_{HIGH} value defines an excess to the operational envelope in a degree of severity that EMBRAER's involvement is necessary to restore the aircraft airworthiness.
- (4) The purpose of this section is to provide the operator with all information to analyze the flight data, determine the required inspections, and restore the aircraft airworthiness. EMBRAER must be involved only if at least one of the situations below occurs:
- The event has occurred at an airspeed higher than the T_{HIGH} value. In this situation, the flight data must be sent to EMBRAER, that will work together with the operator to restore the aircraft airworthiness, or;
 - You find a discrepancy related to envelope exceedance during any phased inspection. In this situation, send the flight data and a report of the discrepancies found to EMBRAER. An EMBRAER disposition is necessary to return the aircraft to service.
- (5) For this event calculation, the parameters below must be selected from the flight data:
- Hours (GMT)
 - Minutes (GMT)
 - Seconds (GMT)
 - Indicated air speed
 - Main Landing Gear Position (Left or Right)
 - Nose Landing Gear Position

**D. Overspeed Landing-Gear Operation - Exceedance Detection with Flight Data available****SUBTASK 200-003-A**

- (1) Identify the flight period when the event occurred in the flight data.
- (2) Examine the value of airspeed and landing gear control-lever position in the interval when the event occurred.
- (3) Make an analysis of the duration of the event as follow:
 - (a) Identify the period of time that the nose or main landing gears were not locked up (in transit or locked down). The values 2 or 3 for the parameters "Nose Landing Gear Position" and "Left/Right Main Landing Gear Position" show the in transit or locked down period.
 - (b) Identify the samples where the airspeed (IAS) was higher than the below values during this period:
 - The V_{LO} (e) value is 265 KIAS.
 - The V_{LO} (r) value is 235 KIAS.
 - The V_{LE} value is 265 KIAS.
- (4) Refer to [Table 601](#) to identify the values for T_{LOW} and T_{HIGH} .

Table 601 - Threshold Values

ENVELOPE (V_{LO} , V_{LE})	THRESHOLD REFERENCE PA- RAMETER	THRESHOLD VALUE
235 KIAS	T_{LOW}	265 KIAS
	T_{HIGH}	280 KIAS
265 KIAS ^[1]	T_{HIGH}	280 KIAS

[1] The maximum airspeed limit is 280 KIAS. For V_{LO} (e) and V_{LE} there is no T_{LOW} .

- (5) Refer to [Table 602](#) for the necessary actions after the event.



Table 602 - Event Severity and Actions

LG POSITION	IAS (Higher Re- corded Sample)	REQUIRED ACTIONS
• In transit or; • Down and locked	Below T _{LOW}	The aircraft can return to service if the flight data shows that the event occurred only once during the flight cycle and the event did not last more than 20 seconds. If at least one of those conditions was not respected, the event must be treated as an event between T _{LOW} and T _{HIGH} .
	Between T _{LOW} and T _{HIGH}	1. Do the Phase-I inspection before the next flight.
		2. If no discrepancy is found during Phase-I inspection, dispatch the aircraft for 10 FC. If you find any discrepancy, contact EMBRAER.
		3. Do the Phase-II inspection during the deferrable period.
	Above T _{HIGH}	4. If no discrepancy is found during Phase-II inspection, return the aircraft to service. If you find any discrepancy, contact EMBRAER.
		5. Send the flight data and the inspection results to EMBRAER if any discrepancy is found during the inspection Phases I or II.
	Above T _{HIGH}	1. Do the Phase-II inspection.
		2. Send the flight data and the inspection results to EMBRAER. Work together with EMBRAER for the necessary actions.

E. Overspeed Landing-Gear Operation - Exceedance Detection with Pilot Crew Report

SUBTASK 200-006-A

(1) The airspeed (IAS) reported by crew was higher than the below values:

- The V_{LO} (e) value is 265 KIAS.
- The V_{LO} (r) value is 235 KIAS.
- The V_{LE} value is 265 KIAS.

(a) The reported event:

- Occurred only once during the flight cycle;
- Below T_{LOW}.
- Did not last more than 20 seconds.

1 If all conditions were respected, the aircraft can return to service.



- 2 If at least one of those conditions were not respected do the procedure for between T_{LOW} and T_{HIGH} . Refer to [Table 602](#).

F. Job Set-Up

WARNING: MAKE SURE THAT THE AIRCRAFT IS IN A SAFE CONDITION BEFORE YOU DO THE MAINTENANCE PROCEDURES. THIS IS TO PREVENT INJURY TO PERSONS AND/OR DAMAGE TO THE EQUIPMENT.

SUBTASK 860-004-A

- (1) Do the procedure to make the aircraft safe for maintenance of the landing gear (AMM TASK 32-00-00-860-801-A/200).
- (2) Open the NLG doors (AMM TASK 32-22-02-860-801-A/200).

G. Overspeed Landing-Gear Operation - Phase I Inspection

SUBTASK 220-001-A

- (1) Examine the landing-gear doors for general misalignment.
- (2) Examine the mechanism rods of all doors for signs of bends.
- (3) Examine all door attachments to the fuselage and wing structures for backlash.
- (4) Examine all hinges for backlash.
- (5) Examine the wheelwell skin and structure for signs of cracks and distortion.
NOTE: Pay special attention to the region surrounding the trunnion and pintle pin attachments.
- (6) Examine the lock stays for signs of bends or distortion.
- (7) Examine the downlock actuators and the extension/retraction actuators for signs of leakage.
- (8) Do a check of the parallelism between the proximity sensor heads.

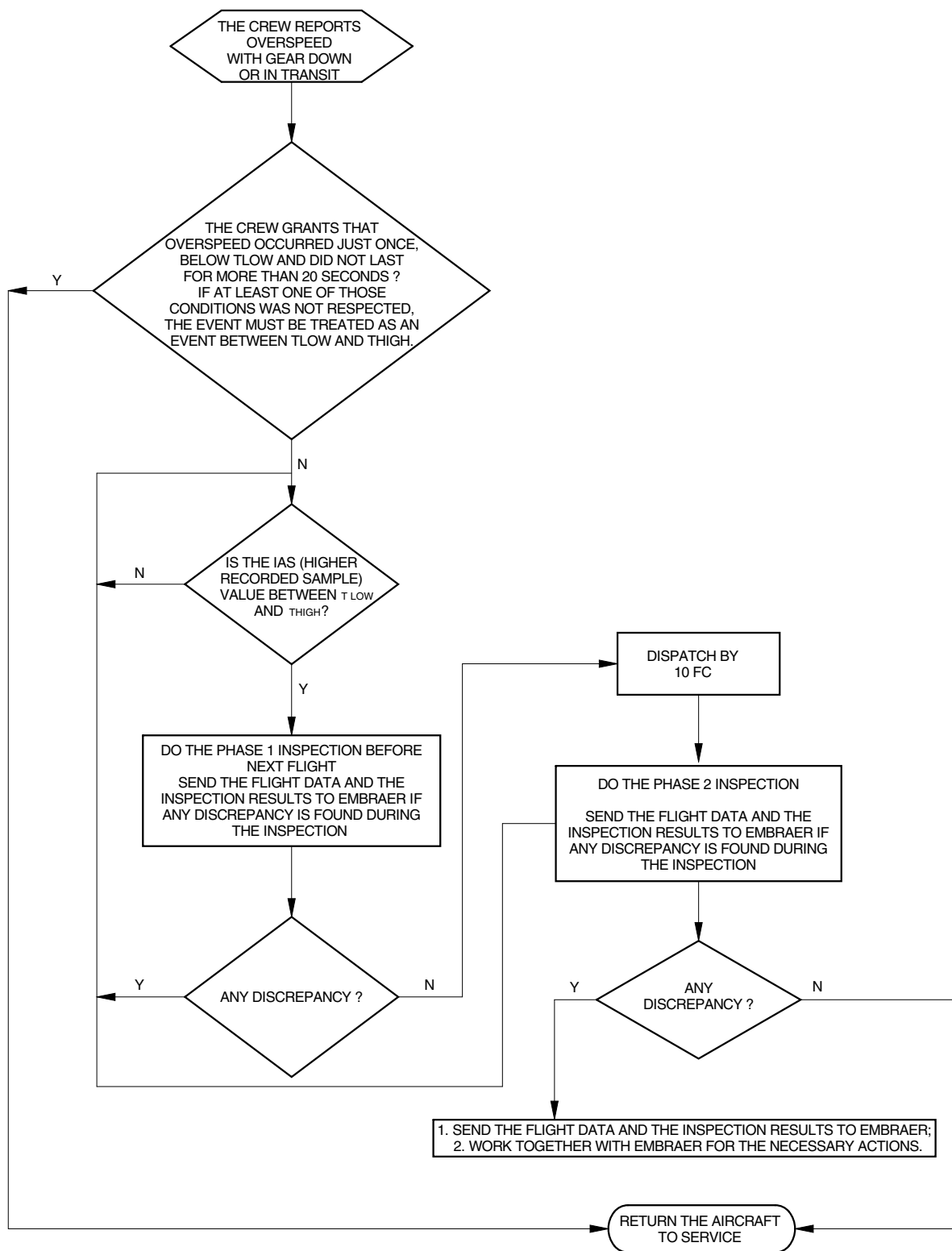
H. Overspeed Landing-Gear Operation - Phase II Inspection

SUBTASK 220-005-A

- (1) Make sure you finished the Phase I Inspection before you start the Phase II inspection.
- (2) Do the these tasks to adjust the gaps and steps:
 - AMM TASK 32-12-00-820-801-A/200
 - AMM TASK 32-22-01-820-801-A/200
 - AMM TASK 32-22-03-820-801-A/200
- (3) Do the these operational tests:
 - AMM TASK 32-33-00-710-801-A/500
 - AMM TASK 32-34-00-710-801-A/500

**I. Job Close-Up****SUBTASK 860-002-A**

- (1) Put the aircraft back to its initial condition (AMM TASK 32-00-00-860-802-A/200).
- (2) Close the NLG doors (AMM TASK 32-22-02-860-802-A/200).
- (3) Remove all tools, equipment, and unwanted materials from the work area.

**EFFECTIVITY: ALL****Overspeed Landing-Gear Operation - Exceedance Detection****Figure 601**

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